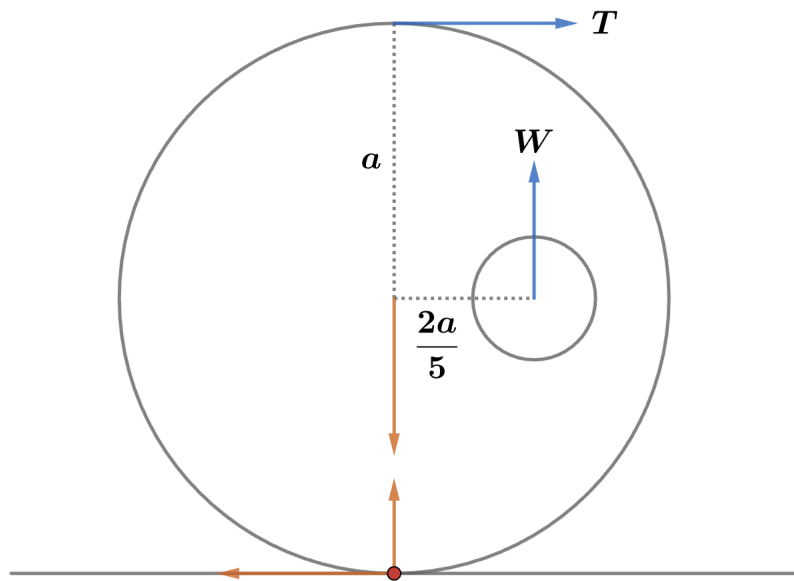


2012 F=ma Exam: Problem 14

Kevin S. Huang

By the principle of superposition, drilling a hole is equivalent to superimposing negative mass at the same location. Thus, we consider the original cylinder with a negative mass cylinder of weight $W = 80\text{ N} - 65\text{ N} = 15\text{ N}$ added to the same location that would have been drilled.



We choose the contact point between the cylinder and the ground as our pivot point and balance torques. Since the normal force, friction force, and weight of the original cylinder all intersect the pivot point, they don't contribute to the torque. We are left with

$$T(2a) = W \left(\frac{2a}{5} \right)$$

where T is the external force applied and W is the weight (pointing upwards) of the negative mass cylinder. We have

$$T = \frac{W}{5} = \frac{15\text{ N}}{5} = 3\text{ N}$$

so the (closest) answer is A.